

The University of the South Pacific

IS333 Project Management

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Assignment 2

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Introduction

The Colonial War Memorial (CWM) Hospital based in Suva, Fiji is the largest public hospital in the country, providing healthcare services to the people of Fiji for a long period of time. Some of the services offered by CWM hospital include emergency care, outpatient services, inpatient services, surgeries, and in-house pharmaceutical services. Currently, CWM hospital is heavily relying on manual operations and is equipped with a partial digital legacy system across its various departments. As a result of this unintegrated system, CWM hospital faces major inefficiencies and challenges in its day-to-day operations, which may also lead to patients being unsatisfied by the services offered by CWM hospital.

In order to solve the current problems faced by CWM hospital, the implementation of a hospital management system is a necessity. The integration and implementation of the hospital management system within the CWM hospital will allow the organization to mitigate operational challenges and inefficiencies caused by the manual processes. The hospital management system will also allow CWM hospital to perform its day-to-day operations such as creating and accessing patient records, scheduling of patient appointments and surgeries for the day and generation of patient and hospital reports for decision making in a timely manner, achieving user satisfaction and overall operational efficiency.

The hospital management system for CWM Hospital would be implemented via the application of Scrum methodology whereby the project team members will follow a continuous cycle of planning, executing and evaluating throughout the development of the hospital management system.

The Scrum methodology will also ensure a focus on the minimum viable product (MVP), which is the simplest working version of the hospital management system for CWM Hospital, that will allow the project team members to receive feedback from the end users with very little effort in order to verify or validate the team ideas (Raj, 2024). Moreover, the Scrum methodology offers multiple agile project management practices such as sprint planning, daily scrums, development work, sprint review and sprint retrospective to ensure close collaboration between the project team members and a high level of user involvement in terms of the development of user stories which will ensure that the needs of the end users of the hospital management system are well captured.

Additionally, key activities like sprint planning will assist the project team to identify the deliverables of the health management system during the sprint and how to achieve those deliverables (West, 2024) while backlog prioritization will consist of a set of items such as user stories and features of the health management system typically ordered from the most important to the least important item and this will ensure that the development team delivers the most important item first (Radigan, 2024). All in all, leading to the successful development of the system itself and achieving user satisfaction.

Project Charter

Project Details

Title	Implementation of Hospital Management System for Colonial War Memorial Hospital
Version	1.0.0

Scope and Objectives

This project aims to successfully implement and integrate hospital management system for the Colonial War Memorial (CWM) Hospital. The implementation of the hospital management system for CWM Hospital will be managed via the application of the Scrum methodology. The objective of this project is to develop a fully digitized hospital management system that would allow CWM Hospital to significantly shift its reliance from a manual to a much more advanced and automated day to day processing, reducing the daily operational inefficiencies and challenges that CWM Hospital currently faces.

Overview

Fiji's largest and oldest hospital, the Colonial War Memorial (CWM) Hospital has been serving the people of Fiji for decades as a central referral medium. Based in Suva, Fiji, CWM Hospital offers a core range of medical services such as emergency, outpatient and inpatient services, together with surgeries and pharmaceutical services. The implementation of the hospital management system will play a vital role in improving the day to day services being offered by the CWM Hospital in terms of handling thousands of patient records in an efficient manner, creating a well-organized and a timely schedule of patient appointments, staff schedule and surgeries for the day, together with allowing for a quick and efficient manner of generating hospital and patient reports for evaluation and decision making.

General Objectives

1. Enhancing patient and staff scheduling across the various departments.
2. Creating an improved hospital service schedule for the day.
3. Real time updating of the pharmacy stock.
4. Improve patient and hospital related report generation services for decision making and analytics.
5. Staff training to ensure data integrity and adherence to privacy regulations.

Specific Objectives

1. Enhancing patient and staff scheduling across the various departments.
 - a) Automated scheduling of patient appointments in outpatient and inpatient departments for the day.
 - b) Automated scheduling of hospital staff, for instance, scheduling of doctors and nurses to the respective departments along with the patient list for the day.
2. Creating an improved hospital service schedule for the day.

- a) Automated scheduling of medical services for the day, for instance, the scheduling of the surgeries to be performed for a particular day, along with the operating theatre details and the list of hospital staff to be involved with it.
3. Real time updating of the hospital pharmacy stock.
 - a) The system will maintain an up-to-date list of pharmacy items in stock.
 - b) Ensure each stock item quantity will be updated in real time upon either being issued to a patient or upon the arrival of new stock.
 - c) Updated pharmacy stock list will be automatically sent to the stock controller for evaluation.
4. Improve patient and hospital related report generation services for decision making and analytics.
 - a) Automated generation of patient medical reports for further evaluation in terms of previous medical histories and current medical conditions to determine the required treatment.
 - b) Automated generation of hospital reports, for instance, financial reports, equipment related reports and medical staff reports such as the number of leaves taken and hours worked for decision making and analytical purposes.
5. Staff training to ensure data integrity and adherence to privacy regulations.
 - a) Training the hospital staff on the core system security aspects such as Role Based Access Control (RBAC) and the use of Multifactor Authentication (MFA) to achieve and maintain data integrity and prevent unauthorized access to the patient and hospital records.
 - b) Educating the hospital staff on the sensitivity of the information being stored within the system to prevent privacy regulation breaches.

Conditions and Constraints

- The hospital management system must be implemented at CWM Hospital within the allocated budget of \$1,000,000.
- The hospital management system must be successfully implemented at CWM Hospital within the timeframe of 12 months.
- The implementation of the hospital management system will follow agile practices based on the Scrum framework.
- The development of the hospital management system for CWM Hospital will adhere to strict backlog prioritization practices, ensuring to deliver the minimum viable product (MVP).
- The hospital management system must be well complied with the rules and regulations of the modern health sector, ensuring patient records are well protected.

Project Organization

The key project team members are:

1. Project Manager – Vishant Kumar
2. Scrum Master – Jason Prasad
3. Software Developers – Anav Chand and Rohan Nandan

4. QA Engineer – Zhixian Chen
5. Business Analyst – James Singh
6. UI/UX Designer – Jai Prasad
7. DevOps Engineer – Jonathan Lal
8. Tech Lead – Steven Shay

Project Manager Responsibilities

1. **Activity and Resource Planning** – The Project Manager must create a viable plan of the project, clearly identifying all the core activities of the project. The Project Manager must also identify the required resources for successfully completing the project.
2. **Managing Risks** – The Project Manager must be responsible for handling the risks associated with the project in an effective and efficient manner.
3. **Time Management** – The Project Manager must ensure that milestones are achieved on the planned schedule, and the overall project is completed within its allocated timeframe, which is 12 months in this case.
4. **Communication and Team Management** – The Project Manager is required to have excellent communication skills in order to communicate with the project and the stakeholders effectively. The Project Manager must also delegate tasks to the project team members and conduct regular meetings to obtain feedback.
5. **Budget** - The allocated budget for this project is \$1,000,000 and the Project Manager is responsible for ensuring that the project does not exceed the allocated budget.
6. **Status Updates** – The Project Manager must provide the higher authorities with the project status updates, typically on a monthly basis.
7. **Managing Change** – The Project Manager must ensure to manage the changes to the project effectively, ensuring not to exceed the allocated budget and schedule.
8. **Maintenance** - The Project Manager must be responsible for creating methodologies in terms of providing support for the final product that has been delivered to the client.

Authority

The project manager has all authority in terms of identifying the necessary tasks and resources that will be required to aid in the completion of the project. Upon the occurrence of resource conflicts, relevant parties like the sponsor and disciplinary VPs will resolve the conflicts.

Approvals

Project Manager: _____

Resource Manager: _____

Compliance Counsel: _____

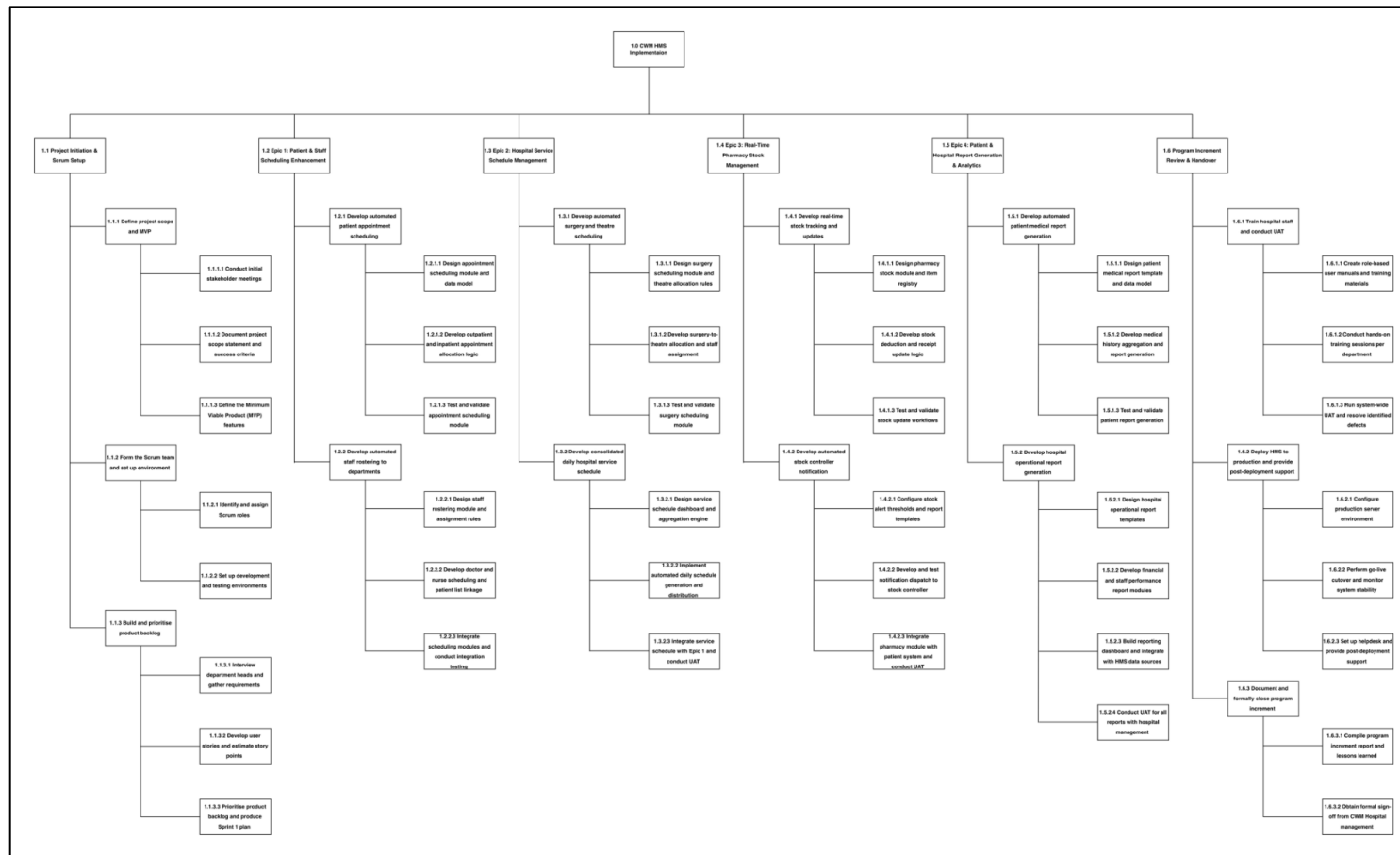
Chief Financial Officer: _____

President: _____

Project Schedule

Work Breakdown Structure

NOTE: If the diagram is hard to interpret, click [here](#) to view the diagram uploaded to OneDrive.



Epic Usage Over Sprints

This project is organized around epics rather than sprints because epics provide the hierarchical decomposition of each major module into project deliverables (user stories and tasks). In contrast, sprints are timeboxed iterations used to execute and manage work across multiple epics, rather than deliverables themselves (Scrum Alliance, n.d.). Furthermore, this approach is more suitable for the project as the MVP is planned over a 12-month development period therefore multiple sprints would occur in a single epic. Additionally, structuring WBS around epics would enable business and feature objectives to be monitored independently of sprint timelines. To conclude, organizing the WBS using epics provides a more long-term and structured view of the project scope and deliverables.

WBS Dictionary

Level	WBS Code	Element name	Definition
1	1.0	CWM HMS Implementation	Complete 12-month program to design, develop, test, and deploy a Hospital Management System for CWM Hospital covering scheduling, pharmacy stock management, and reporting capabilities.
2	1.1	Project Initiation & Scrum Setup	Project is formally started, scope and MVP defined, Scrum team assembled, and a prioritized product backlog produced ready for Sprint 1 planning.
3	1.1.1	Define project scope and MVP	Define the HMS project goals, scope, MVP features, and success criteria with key hospital stakeholders.
4	1.1.1.1	Conduct initial stakeholder meetings	Meet with hospital department heads to identify priorities and expectations for the HMS.
4	1.1.1.2	Document project scope statement and success criteria	Write the formal scope statement outlining what is and is not included in the 12-month program increment.
4	1.1.1.3	Define the Minimum Viable Product (MVP) features	Identify and agree on the full feature set the HMS must deliver across all four Epics by program increment completion
3	1.1.2	Form the Scrum team and set up environment	Assemble the project team with all Scrum roles assigned and the development environment configured and ready.
4	1.1.2.1	Identify and assign Scrum roles	Assign the Product Owner, Scrum Master, and Developer roles to team members.
4	1.1.2.2	Set up development and testing environments	Configure development, testing, and staging server environments and version control repository.

3	1.1.3	Build and prioritize product backlog	Gather requirements from all hospital departments, convert them into user stories, and produce a prioritized, story-pointed backlog ready for sprint planning.
4	1.1.3.1	Interview department heads and gather requirements	Conduct structured interviews with Pharmacy, Scheduling, and Reporting department heads to capture all functional needs.
4	1.1.3.2	Develop user stories and estimate story points	Convert gathered requirements into user stories, estimate story points, and add items to the product backlog.
4	1.1.3.3	Prioritize product backlog and produce Sprint 1 plan	Work with stakeholders to rank backlog items by business value and generate the Sprint 1 sprint backlog.
2	1.2	Epic 1: Patient & Staff Scheduling Enhancement	Delivers automated scheduling of patient appointments and hospital staff across all departments, providing a real-time, conflict-free daily schedule for patients and clinical staff.
3	1.2.1	Develop automated patient appointment scheduling	Build the module that automatically schedules outpatient and inpatient appointments and generates daily patient lists without manual input.
4	1.2.1.1	Design appointment scheduling module and data model	Produce UI wireframes and define the database schema for the patient appointment scheduling module.
4	1.2.1.2	Develop outpatient and inpatient appointment allocation logic	Build the logic of automated assignment of patient appointments to available slots across outpatient and inpatient departments.
4	1.2.1.3	Test and validate appointment scheduling module	Execute unit tests covering appointment allocation scenarios and conflicts
3	1.2.2	Develop automated staff rostering to departments	Build the module that automatically assigns doctors and nurses to their departments with linked patient lists for the day.
4	1.2.2.1	Design staff rostering module and assignment rules	Define rostering rules, department assignment logic, shift constraints, and the module data model.
4	1.2.2.2	Develop doctor and nurse scheduling and patient list linkage	Build the logic of automated assignment of doctors and nurses to departments and link each staff schedule to the corresponding daily patient list.
4	1.2.2.3	Integrate scheduling modules and conduct integration testing	Connect patient appointment and staff rostering modules via shared APIs and execute integration tests to confirm real-time sync and conflict detection.
2	1.3	Epic 2: Hospital Service Schedule Management	Delivers automated daily scheduling of hospital services including surgery and theatre assignments, aggregated into a single hospital-wide service schedule.
3	1.3.1	Develop automated surgery and theatre scheduling	Build the module that automatically schedules surgeries for the day, assigns operating theatres, and links the confirmed surgical team to each procedure.
4	1.3.1.1	Design surgery scheduling module and theatre allocation rules	Define surgery scheduling rules, theatre capacity constraints, and the module data model.

4	1.3.1.2	Develop surgery-to-theatre allocation and staff assignment	Build logic to automatically assign scheduled surgeries to available theatres and link each surgery to its confirmed surgical team list.
4	1.3.1.3	Test and validate surgery scheduling module	Execute unit tests covering theatre allocation, surgical staff assignment, and scheduling conflict rules.
3	1.3.2	Develop consolidated daily hospital service schedule	Build the system that aggregates all department service data into a single daily schedule and distributes it automatically to relevant departments.
4	1.3.2.1	Design service schedule dashboard and aggregation engine	Design the daily hospital service schedule dashboard and build the aggregation engine that pulls service data from all department modules.
4	1.3.2.2	Implement automated daily schedule generation and distribution	Configure automated generation of the daily schedule and its distribution to relevant hospital departments each morning.
4	1.3.2.3	Integrate service schedule with Epic 1 and conduct UAT	Connect the service schedule module to patient and staff scheduling from Epic 1 and conduct user acceptance testing with department heads.
2	1.4	Epic 3: Real-Time Pharmacy Stock Management	Delivers a live pharmacy inventory system that updates stock quantities upon patient issuance or new stock receipt and automatically notifies the stock controller for evaluation.
3	1.4.1	Develop real-time stock tracking and updates	Build the module that updates pharmacy stock quantities in real time whenever items are dispensed to a patient or new stock arrives.
4	1.4.1.1	Design pharmacy stock module and item registry	Define the stock item registry, data model, and module UI for the pharmacy stock management system.
4	1.4.1.2	Develop stock deduction and receipt update logic	Build logic to instantly deduct stock when items are dispensed to a patient and increment stock when new deliveries are recorded.
4	1.4.1.3	Test and validate stock update workflows	Execute unit tests covering deduction, receipt, concurrent update scenarios, and inventory accuracy.
3	1.4.2	Develop automated stock controller notification	Build the automated reporting and notification system that sends updated stock reports to the stock controller whenever thresholds are reached.
4	1.4.2.1	Configure stock alert thresholds and report templates	Define minimum stock level thresholds per item category and design the automated stock summary report template.
4	1.4.2.2	Develop and test notification dispatch to stock controller	Build and test the notification service that sends stock reports to the stock controller upon threshold breach or stock movement events.
4	1.4.2.3	Integrate pharmacy module with patient system and conduct UAT	Link stock issuance records to verified patient encounters in the HMS and conduct UAT with pharmacy and stock control staff.

2	1.5	Epic 4: Patient & Hospital Report Generation & Analytics	Delivers automated generation of patient medical reports and hospital operational reports to support clinical decision-making and hospital management analytics.
3	1.5.1	Develop automated patient medical report generation	Build the module that automatically generates patient medical reports from medical history and current condition data to support clinical decision-making.
4	1.5.1.1	Design patient medical report template and data model	Define the structure, data fields, and format of the automated patient medical report including history and current condition sections.
4	1.5.1.2	Develop medical history aggregation and report generation	Build the engine that compiles a patient's full medical history and current treatment data into the automated report.
4	1.5.1.3	Test and validate patient report generation	Execute unit tests validating report accuracy, completeness, and correct formatting across patient data scenarios.
3	1.5.2	Develop hospital operational report generation	Build the module that automatically generates financial, equipment, and medical staff reports for hospital management decision-making and analytics.
4	1.5.2.1	Design hospital operational report templates	Define templates for financial summaries, equipment status reports, and staff performance reports including leave counts and hours worked.
4	1.5.2.2	Develop financial and staff performance report modules	Build automated generation of hospital financial reports from billing data and staff performance reports from scheduling and HR data.
4	1.5.2.3	Build reporting dashboard and integrate with HMS data sources	Design the central reporting dashboard with role-based access and connect the reporting engine to all HMS modules including scheduling, pharmacy, and billing.
4	1.5.2.4	Conduct UAT for all reports with hospital management	Validate all patient and operational reports with hospital management and department heads and obtain sign-off.
2	1.6	Program Increment Review & Handover	Concludes the 12-month program increment through system-wide testing, staff training, production deployment, and formal handover to establish the foundation for the next program increment.
3	1.6.1	Train hospital staff and conduct UAT	Deliver hands-on training to all staff who will use the HMS and conduct final system-wide user acceptance testing before go-live.
4	1.6.1.1	Create role-based user manuals and training materials	Write department-specific guides and produce quick-reference materials and tutorials for common HMS tasks.
4	1.6.1.2	Conduct hands-on training sessions per department	Run structured training workshops for each department using the live HMS environment.
4	1.6.1.3	Run system-wide UAT and resolve identified defects	Execute user acceptance testing across all four epics with representative hospital staff, log defects, and resolve before production deployment.

3	1.6.2	Deploy HMS to production and provide post-deployment support	Launch the HMS to the live production environment across all CWM Hospital departments and provide ongoing technical support after go-live.
4	1.6.2.1	Configure production server environment	Set up and harden the live server infrastructure that will host the HMS in production.
4	1.6.2.2	Perform go-live cutover and monitor system stability	Switch CWM Hospital operations to the new HMS and closely monitor system health and performance in the immediate post-go-live period.
4	1.6.2.3	Set up helpdesk and provide post-deployment support	Establish a support channel for staff to log HMS issues and provide ongoing bug resolution and performance monitoring after deployment.
3	1.6.3	Document and formally close program increment	Produce a complete end-of-increment report, lessons learned and obtain formal sign-off from CWM Hospital management to close the current program increment.
4	1.6.3.1	Compile program increment report and lessons learned	Write a comprehensive end-of-increment report covering achievements, issues encountered, lessons learned, and recommendations for the next increment.
4	1.6.3.2	Obtain formal sign-off from CWM Hospital management	Secure written approval from hospital leadership confirming the program increment has been delivered successfully and the system is accepted.

Key

Symbol	Meaning
	Deliverable

Activity Table

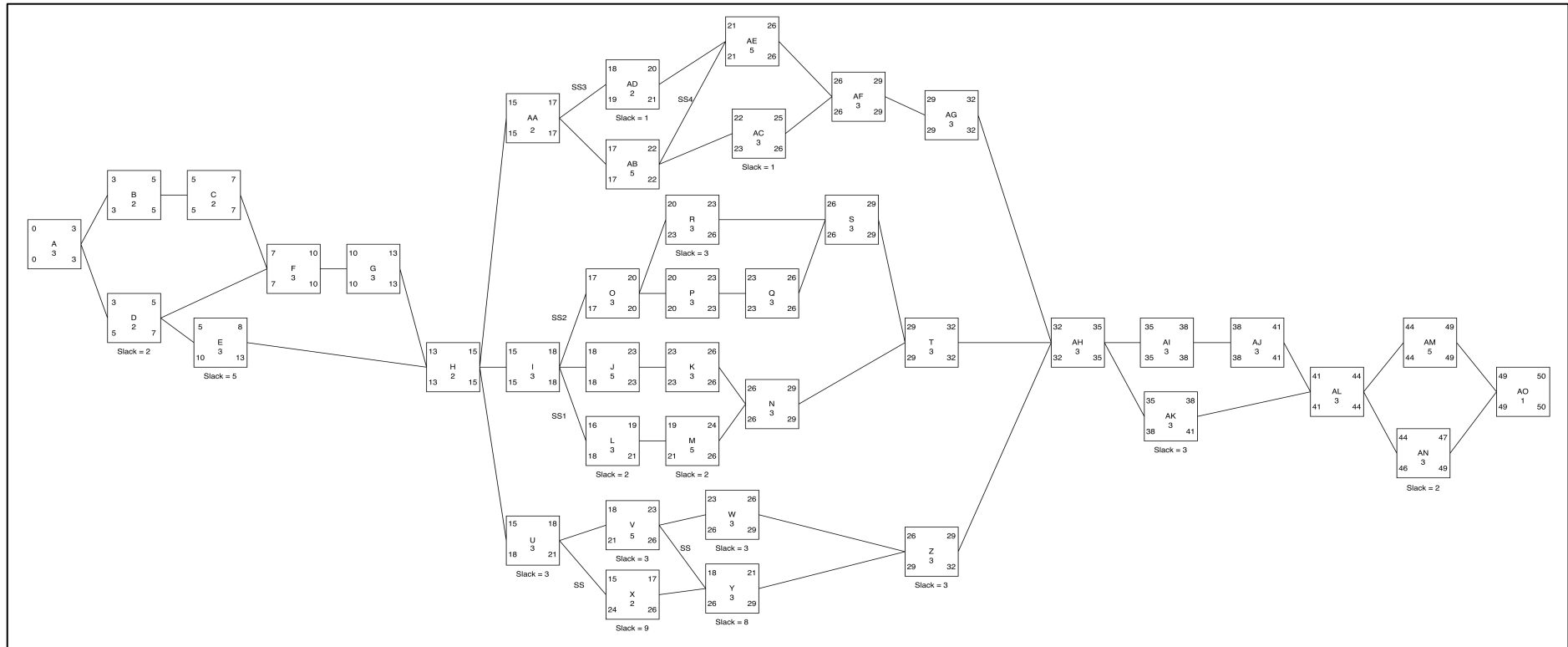
Activity	Activity Name	Optimistic (a)	Most Likely (m)	Pessimistic (b)	Time Estimate (a+4m+b)/6	Duration (Weeks)	Predecessors
A	Conduct Initial Stakeholder Meetings	2	3	4	3.00	3	—
B	Document Project Scope Statement & Success Criteria	2	2	4	2.33	2	A
C	Define MVP Features	2	2	4	2.33	2	B
D	Identify and Assign Scrum Roles	2	2	4	2.33	2	A
E	Set Up Development & Testing Environments	2	3	4	3.00	3	D
F	Interview Department Heads & Gather Requirements	2	3	6	3.33	3	C, D
G	Develop User Stories & Estimate Story Points	2	3	4	3.00	3	F
H	Prioritize Product Backlog & Produce Sprint 1 Plan	2	2	4	2.33	2	G, E
I	Design Appointment Scheduling Module & Data Model	2	3	4	3.00	3	H
J	Develop Outpatient & Inpatient Appointment Allocation	3	5	8	5.17	5	I
K	Test & Validate Appointment Scheduling Module	2	3	4	3.00	3	J
L	Design Staff Rostering Module & Assignment Rules	2	3	4	3.00	3	I (SS1)
M	Develop Doctor & Nurse Scheduling & Patient List Linkage	3	5	8	5.17	5	L
N	Integrate Scheduling Modules & Conduct Integration Testing	2	3	6	3.33	3	K, M
O	Design Surgery Scheduling Module & Theatre Allocation Rules	2	3	4	3.00	3	I (SS2)
P	Develop Surgery-to-Theatre Allocation & Staff Assignment	2	3	5	3.17	3	O

Q	Test & Validate Surgery Scheduling Module	2	3	4	3.00	3	P
R	Design Service Schedule Dashboard & Aggregation Engine	2	3	4	3.00	3	O
S	Implement Automated Daily Schedule Generation & Distribution	2	3	4	3.00	3	R, Q
T	Integrate Service Schedule with Epic 1 & Conduct UAT	2	3	6	3.33	3	S, N
U	Design Pharmacy Stock Module & Item Registry	2	3	4	3.00	3	H
V	Develop Stock Deduction & Receipt Update Logic	3	5	7	5.00	5	U
W	Test & Validate Stock Update Workflows	2	3	4	3.00	3	V
X	Configure Stock Alert Thresholds & Report Templates	2	2	4	2.33	2	U (SS)
Y	Develop & Test Notification Dispatch to Stock Controller	2	3	4	3.00	3	V (SS), X
Z	Integrate Pharmacy Module with Patient System & UAT	2	3	4	3.00	3	W, Y
AA	Design Patient Medical Report Template & Data Model	2	2	4	2.33	2	H
AB	Develop Medical History Aggregation & Report Generation	3	5	7	5.00	5	AA
AC	Test & Validate Patient Report Generation	2	3	4	3.00	3	AB
AD	Design Hospital Operational Report Templates	2	2	4	2.33	2	AA (SS3)
AE	Develop Financial & Staff Performance Report Modules	3	5	7	5.00	5	AB (SS4), AD
AF	Build Reporting Dashboard & Integrate with HMS Data Sources	2	3	6	3.33	3	AE, AC
AG	Conduct UAT for All Reports with Hospital Management	2	3	4	3.00	3	AF
AH	Create Role-Based User Manuals & Training Materials	2	3	4	3.00	3	T, Z, AG
AI	Conduct Hands-On Training Sessions per Department	2	3	4	3.00	3	AH

AJ	Run System-Wide UAT & Resolve Identified Defects	2	3	6	3.33	3	AI
AK	Configure Production Server Environment	2	3	4	3.00	3	AH
AL	Perform Go-Live Cutover & Monitor System Stability	2	3	4	3.00	3	AJ, AK
AM	Set Up Helpdesk & Provide Post-Deployment Support	3	5	7	5.00	5	AL
AN	Compile Program Increment Report & Lessons Learned	2	3	4	3.00	3	AL
AO	Obtain Formal Sign-Off from CWM Hospital Management	1	1	2	1.17	1	AM, AN

AON Diagram

NOTE: The AON diagram can also be accessed on OneDrive by clicking [here](#) if it is difficult to interpret.



Project Duration: 50 weeks

Critical Paths:

A(3) → B(2) → C(2) → F(3) → G(3) → H(2) → I(3) → J(5) → K(3) → N(3) → T(3) → AH(3) → AI(3) → AJ(3) → AL(3) → AM(5) → AO(1) = 50

A(3) → B(2) → C(2) → F(3) → G(3) → H(2) → I(2) → O(3) → P(3) → Q(3) → S(3) → T(3) → AH(3) → AI(3) → AJ(3) → AL(3) → AM(5) → AO(1) = 50

A(3) → B(2) → C(2) → F(3) → G(3) → H(2) → AA(2) → AB(4) → AE(5) → AF(3) → AG(3) → AH(3) → AI(3) → AJ(3) → AL(3) → AM(5) → AO(1) = 50

Cost Management

Resource Estimation/Detailed Budget (FJD)

WBS Code	Activity	Responsibility	Hours Needed	Pay Rate /hr (\$)	Overhead /hr (\$)	Personal Time Mult.	Human Resource (\$)	Facilities (\$)	Equipment (\$)	Materials (\$)	Total Cost (\$)
1.1 Project Initiation & Scrum Setup											
1.1.1.1	Conduct Initial Stakeholder Meetings	Project Manager / BA	40	\$140.00	\$2.00	1.2	6,816.00	210.00	8.50	65.00	7,099.50
1.1.1.2	Document Project Scope & Success Criteria	Project Manager	40	\$90.00	\$2.00	1.2	4,416.00	144.00	9.50	55.00	4,624.50
1.1.1.3	Define MVP Features	BA / Tech Lead	40	\$202.00	\$2.00	1.2	9,792.00	290.00	14.00	10.00	10,106.00
1.1.2.1	Identify and Assign Scrum Roles	Project Manager / Scrum Master	40	\$95.00	\$2.00	1.2	4,656.00	162.00	10.00	22.00	4,850.00
1.1.2.2	Set Up Dev & Testing Environments	DevOps / Tech Lead	64	\$200.00	\$2.00	1.2	15,513.60	300.00	125.00	450.00	16,388.60
1.1.3.1	Interview Dept Heads & Gather Requirements	BA / Project Manager	56	\$165.00	\$2.00	1.2	11,222.40	264.00	22.00	60.00	11,568.40
1.1.3.2	Develop User Stories & Estimate Story Points	BA / Scrum Master	40	\$130.00	\$2.00	1.2	6,336.00	198.00	10.00	28.00	6,572.00
1.1.3.3	Prioritize Backlog & Produce Initial Release Plan	Scrum Master / BA	40	\$120.00	\$2.00	1.2	5,856.00	183.00	9.00	35.00	6,083.00
Sub-Total — 1.1 Project Initiation			360.00				64,608.00	1,751.00	208.00	725.00	67,292.00
1.2 Epic 1 – Patient & Staff Scheduling Enhancement											
1.2.1.1	Design Appointment Scheduling Module & Data Model	UI/UX / Tech Lead	80	\$292.00	\$2.00	1.2	28,224.00	525.00	55.00	115.00	28,919.00
1.2.1.2	Develop Outpatient & Inpatient Appointment Allocation	Dev / Tech Lead	128	\$335.00	\$2.00	1.2	51,763.20	670.00	72.00	95.00	52,600.20

1.2.1.3	Test & Validate Appointment Scheduling Module	QA / Tech Lead	64	\$200.00	\$2.00	1.2	15,513.60	358.00	35.00	55.00	15,961.60
1.2.2.1	Design Staff Rostering Module & Assignment Rules	UI/UX / BA	80	\$265.00	\$2.00	1.2	25,632.00	476.00	42.00	75.00	26,225.00
1.2.2.2	Develop Doctor & Nurse Scheduling & Patient List Linkage	Dev / Tech Lead	128	\$335.00	\$2.00	1.2	51,763.20	670.00	52.00	100.00	52,585.20
1.2.2.3	Integrate Scheduling Modules & Integration Testing	Dev / QA / DevOps	80	\$280.00	\$2.00	1.2	27,072.00	504.00	115.00	30.00	27,721.00
Sub-Total — 1.2 Epic 1			560.00				199,968.00	3,203.00	371.00	470.00	204,012.00
1.3 Epic 2 – Hospital Service Schedule Management											
1.3.1.1	Design Surgery Scheduling Module & Theatre Rules	UI/UX / BA	64	\$230.00	\$2.00	1.2	17,817.60	415.00	75.00	10.00	18,317.60
1.3.1.2	Develop Surgery-to-Theatre Allocation & Staff Assignment	Dev / Tech Lead	128	\$310.00	\$2.00	1.2	47,923.20	651.00	245.00	8.00	48,827.20
1.3.1.3	Test & Validate Surgery Scheduling Module	QA	64	\$200.00	\$2.00	1.2	15,513.60	358.00	28.00	40.00	15,939.60
1.3.2.1	Design Service Schedule Dashboard & Aggregation Engine	UI/UX / Dev	80	\$270.00	\$2.00	1.2	26,112.00	485.00	48.00	35.00	26,680.00
1.3.2.2	Implement Automated Daily Schedule Generation	Dev / DevOps	104	\$280.00	\$2.00	1.2	35,193.60	551.00	6.50	58.00	35,809.10
1.3.2.3	Integrate Service Schedule & Conduct UAT	Dev / QA / BA	80	\$280.00	\$2.00	1.2	27,072.00	504.00	48.00	28.00	27,652.00
Sub-Total — 1.3 Epic 2			520.00				169,632.00	2,964.00	450.50	179.00	173,225.50
1.4 Epic 3 – Real-Time Pharmacy Stock Management											
1.4.1.1	Design Pharmacy Stock Module & Item Registry	UI/UX / BA	64	\$230.00	\$2.00	1.2	17,817.60	415.00	35.00	32.00	18,299.60
1.4.1.2	Develop Stock Deduction & Receipt Update Logic	Dev / Tech Lead	104	\$310.00	\$2.00	1.2	38,937.60	651.00	48.00	28.00	39,664.60

1.4.1.3	Test & Validate Stock Update Workflows	QA	64	\$200.00	\$2.00	1.2	15,513.60	358.00	12.00	120.00	16,003.60
1.4.2.1	Configure Stock Alert Thresholds & Report Templates	BA / DevOps	56	\$145.00	\$2.00	1.2	9,878.40	232.00	7.50	12.00	10,129.90
1.4.2.2	Develop & Test Notification Dispatch to Stock Controller	Dev / QA	80	\$240.00	\$2.00	1.2	23,232.00	432.00	26.00	25.00	23,715.00
1.4.2.3	Integrate Pharmacy Module & Conduct UAT	Dev / QA / BA	64	\$280.00	\$2.00	1.2	21,657.60	504.00	9.00	40.00	22,210.60
Sub-Total — 1.4 Epic 3			432.00				127,036.80	2,592.00	137.50	257.00	130,023.30
1.5 Epic 4 – Patient & Hospital Report Generation & Analytics											
1.5.1.1	Design Patient Medical Report Template & Data Model	BA / UI/UX	64	\$230.00	\$2.00	1.2	17,817.60	415.00	28.00	36.00	18,296.60
1.5.1.2	Develop Medical History Aggregation & Report Generation	Dev / Tech Lead	104	\$320.00	\$2.00	1.2	40,185.60	651.00	48.00	36.00	40,920.60
1.5.1.3	Test & Validate Patient Report Generation	QA	64	\$200.00	\$2.00	1.2	15,513.60	358.00	18.00	30.00	15,919.60
1.5.2.1	Design Hospital Operational Report Templates	BA / UI/UX	64	\$230.00	\$2.00	1.2	17,817.60	415.00	26.00	32.00	18,290.60
1.5.2.2	Develop Financial & Staff Performance Report Modules	Dev / Tech Lead	104	\$320.00	\$2.00	1.2	40,185.60	651.00	36.00	45.00	40,917.60
1.5.2.3	Build Reporting Dashboard & Integrate with HMS Data Sources	Dev / DevOps	104	\$280.00	\$2.00	1.2	35,193.60	551.00	20.00	115.00	35,879.60
1.5.2.4	Conduct UAT for All Reports with Hospital Management	QA / BA / PM	64	\$280.00	\$2.00	1.2	21,657.60	504.00	18.00	30.00	22,209.60
Sub-Total — 1.5 Epic 4			568.00				188,371.20	3,545.00	194.00	324.00	192,434.20
1.6 Program Increment Review & Handover											
1.6.1.1	Create Role-Based User Manuals & Training Materials	BA / PM	64	\$145.00	\$2.00	1.2	11,289.60	240.00	210.00	20.00	11,759.60

1.6.1.2	Conduct Hands-On Training Sessions per Department	BA / PM	64	\$155.00	\$2.00	1.2	12,057.60	258.00	85.00	105.00	12,505.60
1.6.1.3	Run System-Wide UAT & Resolve Identified Defects	QA / Dev / BA	80	\$270.00	\$2.00	1.2	26,112.00	485.00	24.00	132.00	26,753.00
1.6.2.1	Configure Production Server Environment	DevOps / Tech Lead	64	\$220.00	\$2.00	1.2	17,049.60	395.00	50.00	70.00	17,564.60
1.6.2.2	Perform Go-Live Cutover & Monitor System Stability	DevOps / Tech Lead	80	\$200.00	\$2.00	1.2	19,392.00	360.00	610.00	82.00	20,444.00
1.6.2.3	Set Up Helpdesk & Provide Post-Deployment Support	DevOps	100	\$180.00	\$2.00	1.2	21,840.00	324.00	35.00	90.00	22,289.00
1.6.3.1	Compile Program Increment Report & Lessons Learned	PM / BA	40	\$145.00	\$2.00	1.2	7,056.00	218.00	390.00	45.00	7,709.00
1.6.3.2	Obtain Formal Sign-Off from CWM Hospital Management	PM	24	\$100.00	\$2.00	1.2	2,937.60	144.00	9.00	25.00	3,115.60
Sub-Total — 1.6 Program Increment Review			516.00				117,734.40	2,424.00	1,413.00	569.00	122,140.40
GRAND TOTAL — ALL PHASES			2,956.00				867,350.40	16,479.00	2,774.00	2,524.00	889,127.40
BUDGET SUMMARY											
Total Budget (FJD)								\$1,000,000.00			
Estimated Project Cost (FJD)								\$889,127.40			
Budget Remaining (FJD)								\$110,872.60			
Budget Utilization %								88.9%			

Summary Budget

Summary Budget for CWM HMS Project	
Expense Category	Amount (FJD)
Expenses	
1.1 Project Initiation & Scrum Setup	
Human Resource	\$64,608.00
Facilities	\$1,751.00
Equipment	\$208.00
Materials	\$725.00
Sub Total	\$67,292.00
1.2 Epic 1 – Patient & Staff Scheduling Enhancement	
Human Resource	\$199,968.00
Facilities	\$3,203.00
Equipment	\$371.00
Materials	\$470.00
Sub Total	\$204,012.00
1.3 Epic 2 – Hospital Service Schedule Management	
Human Resource	\$169,632.00
Facilities	\$2,964.00
Equipment	\$450.50
Materials	\$179.00
Sub Total	\$173,225.50
1.4 Epic 3 – Real-Time Pharmacy Stock Management	
Human Resource	\$127,036.80
Facilities	\$2,592.00
Equipment	\$137.50
Materials	\$257.00
Sub Total	\$130,023.30
1.5 Epic 4 – Patient & Hospital Report Generation & Analytics	
Human Resource	\$188,371.20
Facilities	\$3,545.00
Equipment	\$194.00
Materials	\$324.00
Sub Total	\$192,434.20
1.6 Program Increment Review & Handover	

Human Resource	\$117,734.40
Facilities	\$2,424.00
Equipment	\$1,413.00
Materials	\$569.00
Sub Total	\$122,140.40
Total Project Cost	\$889,127.40
Budget Overview	
Total Budget	FJD \$1,000,000.00
Estimated Cost	FJD \$889,127.40
Budget Remaining	FJD \$110,872.60
Budget Utilization	88.9%

Skill Matrix

Skill Category	Skill	Vishant Kumar	Jason Prasad	Anav Chand	Rohan Nandan	Zhixian Chen	James Singh	Jai Prasad	Jonathan Lal	Steven Shay
		Project Manager	Scrum Master	Software Developer	Software Developer	QA Engineer	Business Analyst	UI/UX Designer	DevOps Engineer	Tech Lead
Project & Management	Project Management	✓								
	Stakeholder Coordination	✓	✓							
	Scope Management	✓								
	Change Management	✓					✓			
	Documentation	✓					✓			
Business Analysis & Requirements	Requirements Gathering	✓					✓			✓
	Vendor Evaluation	✓								
	Module Configuration	✓					✓			✓
	ERP Implementation Expertise	✓								✓
	Data Validation					✓	✓			
Software Development	Coding			✓	✓					✓

	Prototyping			✓	✓			✓		
	Database Normalization			✓	✓					✓
	Data Migration			✓	✓				✓	
	Data Integration			✓	✓				✓	
Design	UI Design							✓		
	Multi-Channel Support							✓		
Quality Assurance & Testing	System Testing					✓				✓
	Integration Testing					✓			✓	✓
	User Acceptance Testing					✓	✓			
DevOps & Infrastructure	Technical Support								✓	✓
	Troubleshooting								✓	✓
	System Maintenance								✓	✓
Agile & Scrum	Agile / Scrum Facilitation		✓							
	Sprint Planning	✓	✓							

	Backlog Management	✓	✓							
Compliance & Regulatory	Regulatory and Compliance Management									
Analytics & Reporting	Analytics						✓			
	Business Intelligence Tool Integration						✓			
	Financial System Integration						✓			
Training	Training Program Development						✓			

Needs Matrix

Skill Abbreviations Table

Skill Abbreviations Table	
Skill	Abbreviation
Project Management	LEAD
Stakeholder Coordination	SKTH
Scope Management	STRT
Documentation	COMM
ERP Implementation Expertise	FUNC
Requirements Gathering	REQM
Vendor Evaluation	VNDR
Module Configuration	CONF
Coding	CODE
Prototyping	DESN
UI Design	USXP
Data Validation	DQTY
Regulatory & Compliance Management	SEC
Customer Service Workflows	BPRC
Multi-Channel Support	MCHP
Training Program Development	TRNG
Change Management	CHNG
Analytics	RPTG
Financial System Integration	FINT
BI Tool Integration	DTAN
Technical Support	SUPT
Troubleshooting	PRBL
System Maintenance	SMNT
Database Normalization	DBNM
Data Migration	DMIG
Data Integration	DINT
System Testing	STST
Integration Testing	ITST
User Acceptance Testing	UACC

Needs Inventory

Needs Inventory																													
WBS Code	Activity Name	L E A D	S K T H	S T R E E T	C O M M U N I T Y	F U N C T I O N	R E Q U I R E M E N T	V E N U E	C O N F E R E N C E	D E S I G N	U S E R S T O R Y	D E S I G N	S E R V I C E	B E N E F I T	M E A S U R E	T I M E	C H A R A C T E R	R E S O U R C E	F I N A N C I A L	D I S T R I B U T I O N	S U P P L Y	P R O D U C T	S E R V I C E	D E M O	D I S T R I B U T I O N	S E R V I C E	I T E M	U S E	
1.1.1.1	Conduct Initial Stakeholder Meetings	✓	✓				✓																						
1.1.1.2	Document Project Scope Statement & Success Criteria	✓		✓	✓																								
1.1.1.3	Define the MVP Features	✓	✓	✓			✓																						
1.1.2.1	Identify and Assign Scrum Roles	✓	✓															✓											
1.1.2.2	Set Up Development & Testing Environments								✓												✓		✓						
1.1.3.1	Interview Department Heads & Gather Requirements		✓				✓																						
1.1.3.2	Develop User Stories & Estimate Story Points						✓			✓																			
1.1.3.3	Prioritize Product Backlog & Produce Sprint 1 Plan	✓	✓				✓																						
1.2.1.1	Design Appointment Scheduling Module & Data Model									✓	✓														✓				
1.2.1.2	Develop Outpatient & Inpatient Appointment Allocation								✓																✓		✓		
1.2.1.3	Test & Validate Appointment Scheduling Module											✓																✓	
1.2.2.1	Design Staff Rostering Module & Assignment Rules						✓			✓	✓																		
1.2.2.2	Develop Doctor & Nurse Scheduling & Patient List Linkage								✓																✓		✓		
1.2.2.3	Integrate Scheduling Modules & Conduct Integration Testing																									✓		✓	
1.3.1.1	Design Surgery Scheduling Module & Theatre Allocation Rules						✓			✓	✓																		
1.3.1.2	Develop Surgery-to-Theatre Allocation & Staff Assignment								✓																✓		✓		
1.3.1.3	Test & Validate Surgery Scheduling Module											✓																✓	
1.3.2.1	Design Service Schedule Dashboard & Aggregation Engine									✓	✓								✓		✓								
1.3.2.2	Implement Automated Daily Schedule Generation & Distribution								✓																	✓			
1.3.2.3	Integrate Service Schedule with Epic 1 & Conduct UAT																									✓		✓	✓
1.4.1.1	Design Pharmacy Stock Module & Item Registry									✓	✓														✓				
1.4.1.2	Develop Stock Deduction & Receipt Update Logic								✓																✓		✓		

Staff Assignment Matrix

WBS Code	Activity	Vishant Kumar	Jason Prasad	Anav Chand	Rohan Nandan	Zhixian Chen	James Singh	Jai Prasad	Jonathan Lal	Steven Shay
		Project Manager	Scrum Master	Software Developer	Software Developer	QA Engineer	Business Analyst	UI/UX Designer	DevOps Engineer	Tech Lead
1.1.1.1	Conduct Initial Stakeholder Meetings	●	●				●			●
1.1.1.2	Document Project Scope Statement & Success Criteria	●	●				●			
1.1.1.3	Define the MVP Features	●	●				●			●
1.1.2.1	Identify and Assign Scrum Roles	●	●							
1.1.2.2	Set Up Development & Testing Environments			●	●				●	●
1.1.3.1	Interview Department Heads & Gather Requirements	●	●				●	●		
1.1.3.2	Develop User Stories & Estimate Story Points		●				●			●
1.1.3.3	Prioritize Product Backlog & Produce Sprint 1 Plan	●	●				●			●
1.2.1.1	Design Appointment Scheduling Module & Data Model			●	●			●		●
1.2.1.2	Develop Outpatient & Inpatient Appointment Allocation			●	●					●
1.2.1.3	Test & Validate Appointment Scheduling Module					●				●
1.2.2.1	Design Staff Rostering Module & Assignment Rules			●	●		●	●		●
1.2.2.2	Develop Doctor & Nurse Scheduling & Patient List Linkage			●	●					●
1.2.2.3	Integrate Scheduling Modules & Conduct Integration Testing			●	●	●			●	●
1.3.1.1	Design Surgery Scheduling Module & Theatre Allocation Rules			●	●		●	●		●
1.3.1.2	Develop Surgery-to-Theatre Allocation & Staff Assignment			●	●					●
1.3.1.3	Test & Validate Surgery Scheduling Module					●				●

1.3.2.1	Design Service Schedule Dashboard & Aggregation Engine			●	●			●	●	●
1.3.2.2	Implement Automated Daily Schedule Generation & Distribution			●	●				●	●
1.3.2.3	Integrate Service Schedule with Epic 1 & Conduct UAT			●	●	●	●			●
1.4.1.1	Design Pharmacy Stock Module & Item Registry			●	●			●		●
1.4.1.2	Develop Stock Deduction & Receipt Update Logic			●	●					●
1.4.1.3	Test & Validate Stock Update Workflows					●				●
1.4.2.1	Configure Stock Alert Thresholds & Report Templates						●		●	●
1.4.2.2	Develop & Test Notification Dispatch to Stock Controller			●	●	●				●
1.4.2.3	Integrate Pharmacy Module with Patient System & UAT			●	●	●	●			●
1.5.1.1	Design Patient Medical Report Template & Data Model			●	●		●	●		●
1.5.1.2	Develop Medical History Aggregation & Report Generation			●	●				●	●
1.5.1.3	Test & Validate Patient Report Generation					●	●			●
1.5.2.1	Design Hospital Operational Report Templates						●	●	●	●
1.5.2.2	Develop Financial & Staff Performance Report Modules			●	●				●	●
1.5.2.3	Build Reporting Dashboard & Integrate with HMS Data Sources			●	●			●	●	●
1.5.2.4	Conduct UAT for All Reports with Hospital Management	●				●	●			●
1.6.1.1	Create Role-Based User Manuals & Training Materials	●	●				●	●		
1.6.1.2	Conduct Hands-On Training Sessions per Department	●	●				●	●		
1.6.1.3	Run System-Wide UAT & Resolve Identified Defects	●				●	●			●
1.6.2.1	Configure Production Server Environment			●	●				●	●

1.6.2.2	Perform Go-Live Cutover & Monitor System Stability	●							●	●
1.6.2.3	Set Up Helpdesk & Provide Post-Deployment Support								●	●
1.6.3.1	Compile Program Increment Report & Lessons Learned	●	●				●			
1.6.3.2	Obtain Formal Sign-Off from CWM Hospital Management	●	●				●			●

Risk Management Plan

Top Three Risks

The top three risks (ranked by high probability and impact) are:

1. Patient Data Breach

CWM Hospital handles a large volume of sensitive patient data on a monthly basis. This data includes medical histories, treatment plans, pharmaceutical records, personal identifiers and much more. The current systems fragmented nature (manual work and legacy systems) makes data confidentiality during transition a challenge. As a result, threats such as unauthorized access, external breach, weak authentication and data exposure during migration can arise.

2. Legacy System Integration Failure

The current hospital currently makes use of a mix of both manual processes and partially digitized legacy systems. This is being used across multiple departments within the hospital, with no form on integration between the systems. The integration process could result in various issues. Firstly, the data format between these systems may have compatibility issues. Secondly, there could be API failures when the HMS tries to sync data. Lastly, data may get corrupted during this synchronization process.

3. Project Budget Overrun

The fixed budget of \$1,000,000 (which is allocated to be spent across a 12-month timeline), may not be enough considering how large this HMS MVP project is. Since there are different independent hospital departments, their involvement throughout the project may lead to scope creep. If requested features get accepted without proper, it could easily result in going over the budget. Similarly, unexpected defects discovered during UAT will require fixing (which requires development time) and can hurt the budget and schedule.

Risk Monitoring

Risk monitoring will be a continuous process that will last the entire project duration of 12 months. The three highest scoring risks will be a part of the project schedule as formal monitoring checkpoints. The budget risk is owned by Vishant Kumar while the other two technical risks are owned by Steven Shay. Both risk managers will provide status updates in team meetings every fortnight. Monitoring will take place both on the sprint level and project management level. At the sprint level, the project manager will review the risk register in each sprint planning session. At the project level, the project manager will perform formal reviews every two weeks. Triggers defined in the risk register will act as an early warning sign. The assigned risk manager will escalate immediately to the project manager when a trigger is observed rather than waiting for the scheduled review.

Risk Register

Rank	Risk	Category	Probability	Impact	Score	Response	Trigger	Owner
1	Patient data breach or unauthorized access	Technical	9	10	90	Minimize	Failed login attempts or unauthorized access alert	Steven Shay
2	Legacy system integration failure	Technical	8	10	80	Minimize	Integration testing has API errors or sync failures	Steven Shay
3	Project budget overrun	Financial	7	9	63	Minimize	Amount spent in a month goes beyond planned budget baseline	Vishant Kumar
4	Key developer leaves	Execution	6	9	54	Minimize	Core developer provides resignation notice	Vishant Kumar
5	Non-compliance with health data regulations	Legal	5	10	50	Minimize	Data handling gaps get flagged during legal reviews	Vishant Kumar
6	Scope creep	Execution	8	7	56	Minimize	More than two unplanned changes approved in a sprint	Vishant Kumar
7	Poor system performance under high load	Technical	7	7	49	Minimize	Response time threshold exceeded during load testing	Steven Shay
8	Vendor software licensing non-compliance	Legal	5	9	45	Transfer	License audit identifies unlicensed software components	Vishant Kumar
9	Stakeholder blockage (slow feedback)	Execution	7	6	42	Minimize	More than 3 business days to get sprint review sign off	Vishant Kumar
10	Data loss during migration from legacy systems	Technical	6	7	42	Minimize	Post migration record count has a mismatch	Steven Shay
11	HSM adoption resistance by staff	Execution	7	5	35	Minimize	Negative UAT feedback and poor train session engagement	Vishant Kumar

12	Third-party vendor delays in software delivery	Financial	6	6	36	Share	Agreed delivery milestone is missed by vendor	Vishant Kumar
13	Change in health sector regulations	Legal	4	8	32	Accept	New health directive issued by Government of Ministry of Health	Vishant Kumar
14	Product environment has insufficient server infrastructure	Technical	5	6	30	Minimize	Capacity shortfall identified during infrastructure assessment	Jonathan Lal
15	Foreign exchange rate fluctuation	Financial	5	5	25	Accept	More than 5% change from baseline on exchange rates (FJD/USD)	Vishant Kumar

Key

Symbol	Meaning
	High Risk
	Medium Risk
	Low Risk

Risk Mitigation

Rank	Risk	Category	Mitigation Technique
1	Patient data breach or unauthorized access	Technical	All HMS accounts must have Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA)
2	Legacy system integration failure	Technical	Conduct a detailed legacy system audit before integration sprints begin.
3	Project budget overrun	Financial	Track expenditure against the baseline budget on a fortnightly basis.
4	Key developer leaves	Execution	Follow standards, document code and share knowledge across all sprints.
5	Non-compliance with health data regulations	Legal	HMS data handling must be reviewed by a compliance consultant against Fiji health regulations.
6	Scope creep	Execution	All new requirements must be submitted via CRF, assessed for impact, and approved by the CCB.
7	Poor system performance under high load	Technical	Load and stress tests must be conducted in a staging/QA environment before production deployment.
8	Vendor software licensing non-compliance	Legal	All third-party licenses must be reviewed with legal counsel before purchase.
9	Stakeholder blockage (slow feedback)	Execution	Allow stakeholders a 3 business day feedback windows and confirm sprint review schedules.
10	Data loss during migration from legacy systems	Technical	All legacy data must be fully backed up before the migration begins.
11	HSM adoption resistance by staff	Execution	UAT must being in early sprints, accommodating users from different departments and roles.
12	Third-party vendor delays in software delivery	Financial	Obligations for delivery milestones and penalties must be included in vendor contracts.
13	Change in health sector regulations	Legal	Regulatory publications from health officials must be monitored throughout the project.
14	Product environment has insufficient server infrastructure	Technical	All required production servers must be provisioned by month 9 after an infrastructure assessment.
15	Foreign exchange rate fluctuation	Financial	Negotiate fixed price contracts with vendors. Else, accept risk as it is a very low score.

Change Management Plan

Change Request Process

The CWM Management System Project makes use of the following step by step process which governs how scope change requests are handled:

1. **Submission:** a complete change request form (CRF) needs to be submitted to the project manager. This may be done by any project stakeholder. Changes should not be discussed verbally, and all requests should be properly documented.
2. **Assessment:** the project manager and team lead conducts a formal impact assessment, which covers the project scope, budget, schedule and quality. This should be done within 5 days of submission, and is documented in the CRF
3. **Review:** the completed CRF and impact assessment are presented to the change control board. They evaluate the request against the projects fixed constraints (\$1M budget, 12-month timeline), and either accept, reject or deter the request.
4. **Integration:** approved changes are added to the product backlog for future sprints. It is prioritized by the product owner and scheduled for the next appropriate sprint. The project manager will update the WBS, schedule and budget.
5. **Implementation:** developers implement the change within the assigned sprint, and it goes through the standard QA and testing process. Upon completion, the project manager marks the change request as closed and notifies relevant stakeholders.

Change Request Form

Below is the Change Request Form template for the CWM Hospital Management System project:

Request Details	
Request Id:	Submission Date:
Requested By:	Role:
Change Description:	
Business Justification:	
Affected Modules:	

Impact Assessment		
Area	Description	Severity
Scope		
Budget		
Schedule		
Quality		

CCB Decision	
Decision Date:	Review Date;
Decision:	
Decision Notes:	

Integration & Implementation	
Sprint:	Assignee:
Baseline Updated:	Completion Date:
Closure Notes:	

Communication Plan

Since this project adopts Scrum, communication, monitoring, and controlling will be managed through regular Scrum events.

To bridge the gap between Project Team and stakeholders, a project status report about report schedule, budget, scope, risks, and overall progress will be delivered to the hospital sponsor and other relevant stakeholders monthly. Scrum stand-up meetings will be conducted daily to synchronize and schedule project progress within the Scrum team. Each sprint will last two weeks, to control and improve the project's progress in sprint planning meetings and review meetings; retrospective meetings will be conducted before and after each sprint.

To ensure the quality of the system, the QA engineering will conduct testing continuously during the entire system development life cycle and before go-live. The QA engineering will provide testing reports and defect reports to team members for correction.

Communication Item	Participants	Deliverable	Frequency	Owner	Method	Purpose
Initial Stakeholder Meeting	Project Manager, Business Analyst, and Hospital Department Heads	Formal Scope Statement, Success Criteria, Initial Product Backlog	Once before development begin	Project Manager/Business Analyst	In-Person Meeting/Meeting Notes	Gather formal requirements, Identify priorities and expectations
Scrum Meeting	Scrum Team	Project Status Update	Daily	Scrum Master	In-Person Meeting/Meeting Notes	Track daily progress, blockers, and sprint tasks

Sprint Planning	Scrum Team, Project Manager, Business Analyst	Sprint Backlog	Fortnightly	Scrum Master	In-Person Meeting/Zoom Meeting	Determine backlog items for the next sprint and define sprint goals
Sprint Review	Scrum Team, Hospital Sponsor, Hospital Department Head	Demo Notes, Stakeholder's Feedback	End of each Sprint	Project Manager / Scrum Master	In-Person Meeting/Zoom Meeting	Demonstrate completed features and collect feedback
Sprint Review Feedback	Hospital Sponsor, Hospital Department Heads, Project Manager	Signed Sprint Acceptance Form, Logged Change Request Form	Within 3 business days after each Sprint Review	Project Manager	In-Person Meeting/Zoom Meeting/Email	Ensure features meet requirements and avoid schedule delay
Sprint Retrospective	Scrum Team	Retrospective Action List	End of each Sprint	Scrum Master	In-Person Meeting/ Zoom Meeting/Meeting Notes	Improve next sprint
Project Status Report	Hospital Sponsor, Senior Stakeholders	Project Status Report	Monthly	Project Manager	Project Status Report	Inform stakeholders about project progress
Risk Review Meeting	Project Manager, Scrum Master, Tech Lead, QA Engineer	Risk Register Update	When major risk occurs	Project Manager	In-Person Meeting/Zoom Meeting/Risk Register Update	Review technical, execution, legal, and financial risks
Change Control Board Review	Project Manager, Involved Stakeholders	Change Request Form and Decision Log	As required	Project Manager	In-Person Meeting/Zoom Meeting	Approve/Reject change requests

Quality and Testing Report	Project Team, Hospital Sponsor, Hospital Department Head	Test Report, Defect Log, UAT Result	Each testing phase and before go-live	QA Engineer	Test Report, Defect Log, UAT Result	Monitor and control the quality of the project
Go-Live Readiness Meeting	All Stakeholders	Go-Live Checklist	Once before Deployment	Project Manager	In-Person Meeting/Zoom Meeting	Confirm training, infrastructure, data, support, and deployment readiness
Departmental Hands-On Training	Business Analyst, Project Manager, Hospital Staff	Role-based User Manual, Training logs, Feedback Forms	Once on Month 12	Business Analyst	In-Person Workshop/User Manual	Ensure actual product delivery and collect structured feedback
Post-Deployment Support	Hospital, Support Team	Helpdesk Tickets, Issue Log	As Required	DevOps Engineer/Tech Lead	Helpdesk Tickets, Issue Log	Track system stability and collect user feedback

Conclusion

With the comprehensive project charter, WBS, risk management, change management and communication plan, the project is well positioned to provide a practical solution to operational and clinical risks created by the Colonial War Memorial Hospital's manual, fragment legacy system. With the help of Hospital Management System, the CWM hospital will be able to deliver safer, faster and more consistent services. This project uses Scrum methodology to support stakeholders' involvement across the entire system development lifecycle and to control the project within the \$1,000,000 FJD budget and 12 months' time limit.

During the project planning process, the team obtained three key insights. Firstly, Patient & Staff Scheduling should be treated as the MVP with highest priority since other modules depend on it, which requires more resources. Secondly, the high-risk score of technical issues of patient data breach and integration failure requires early involvement and strict control. Thirdly, there exists the potential challenge of staff resistance to a new automatic digital data flow, which requires the project team's attention on UAT and staff training.

With focus on MVP, the recommendations are as follows:

The project team should prioritize patient and staff scheduling within the backlog to make it stabilized in advance. Data integrity should be emphasized, and testing should be conducted in time during system development. To address the potential possibility of integration failure and user acceptance failure, the project team should ensure the engagement of hospital stakeholders by Sprint review and allocate more resources to staff training to minimize staff resistance to the new system.

References

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Appendix

Terms of Reference

Group Members

Name	Student ID	Student Email
Anav Chand	S11221203	S11221203@student.usp.ac.fj
Vishant Kumar	S11230430	S11230430@student.usp.ac.fj
Zhixian Chen	S11230686	S11230686@student.usp.ac.fj
Rohan Nandan	S11234883	S11234883@student.usp.ac.fj

Team Purpose

This team is formed to complete IS333 Assignment 2 of Semester 1, 2026. The team will collaboratively develop a comprehensive IT Project Management Plan for the implementation of a Hospital Management System at Colonial War Memorial (CWM) Hospital, applying Scrum methodology and project management knowledge gained throughout the course.

Team Roles

Name	Role
Vishant Kumar	Project Manager
Anav Chand	Software Developer
Rohan Nandan	Software Developer
Zhixian Chen	QA Engineer

Resources

- Assignment 2 Specification: task requirements, deadlines, and marking criteria.
- The Internet and USP Library: research and academic references to support findings.
- Lecture and Course Materials: guidance on Scrum, WBS, risk management, and other PM concepts covered in class.
- Diagramming and Documentation Tools; MS Visio, Draw.io, MS Word, or other approved tools for diagrams, tables, and report preparation.

Governance

All decisions will be discussed collectively with each member's input considered before finalization. The team is committed to fairness, accountability, and timely submission of a high-quality report.

Codes of Conduct

- Communicate openly and honestly regarding progress and challenges.
- Actively participate in meetings and discussions.
- Complete assigned tasks within agreed deadlines.
- Review all work collaboratively before submission.
- Maintain academic integrity and confidentiality throughout.

Responsibility Assignment Matrix

Responsibility Assignment Matrix										
WBS	Activity	Vishant Kumar Project Manager	Jason Prasad Scrum Master	Anav Chand Software Developer	Rohan Nandan Software Developer	Zhixian Chen QA Engineer	James Singh Business Analyst	Jai Prasad UI/UX Designer	Jonathan Lal DevOps Engineer	Steven Shay Tech Lead
1.1.1.1	Conduct Initial Stakeholder Meetings	○	◇				◇			◇
1.1.1.2	Document Project Scope Statement & Success Criteria	○	◇				◇			
1.1.1.3	Define the MVP Features	○	◇	□	□		◇			◇
1.1.2.1	Identify and Assign Scrum Roles	○	○							
1.1.2.2	Set Up Development & Testing Environments			○	○				◇	◇
1.1.3.1	Interview Department Heads & Gather Requirements	◇	◇				○	◇		
1.1.3.2	Develop User Stories & Estimate Story Points		◇				○			◇
1.1.3.3	Prioritize Product Backlog & Produce Sprint 1 Plan	○	○				◇			◇
1.2.1.1	Design Appointment Scheduling Module & Data Model			○	○		●	◇		
1.2.1.2	Develop Outpatient & Inpatient Appointment Allocation			○	○					●
1.2.1.3	Test & Validate Appointment Scheduling Module	□		□	□	●				●
1.2.2.1	Design Staff Rostering Module & Assignment Rules			○	○		◇	◇		●
1.2.2.2	Develop Doctor & Nurse Scheduling & Patient List Linkage			○	○					●
1.2.2.3	Integrate Scheduling Modules & Conduct Integration Testing			○	○	◇			◇	●
1.3.1.1	Design Surgery Scheduling Module & Theatre Allocation Rules			○	○		●	◇		
1.3.1.2	Develop Surgery-to-Theatre Allocation & Staff Assignment			○	○					●

1.3.1.3	Test & Validate Surgery Scheduling Module	□		□	□	●				●
1.3.2.1	Design Service Schedule Dashboard & Aggregation Engine			○	○		●	◇	◇	
1.3.2.2	Implement Automated Daily Schedule Generation & Distribution			○	○				◇	●
1.3.2.3	Integrate Service Schedule with Epic 1 & Conduct UAT			○	○	◇	◇			●
1.4.1.1	Design Pharmacy Stock Module & Item Registry			○	○		●	◇		
1.4.1.2	Develop Stock Deduction & Receipt Update Logic			○	○					●
1.4.1.3	Test & Validate Stock Update Workflows	□		□	□	●				
1.4.2.1	Configure Stock Alert Thresholds & Report Templates			○	○		◇		◇	●
1.4.2.2	Develop & Test Notification Dispatch to Stock Controller			○	○	◇				●
1.4.2.3	Integrate Pharmacy Module with Patient System & UAT			○	○	◇	◇			●
1.5.1.1	Design Patient Medical Report Template & Data Model			○	○		●	◇		
1.5.1.2	Develop Medical History Aggregation & Report Generation			○	○				◇	●
1.5.1.3	Test & Validate Patient Report Generation			○		●	◇			
1.5.2.1	Design Hospital Operational Report Templates						●	○	◇	
1.5.2.2	Develop Financial & Staff Performance Report Modules			○	○				◇	●
1.5.2.3	Build Reporting Dashboard & Integrate with HMS Data Sources			○	○			◇	◇	●
1.5.2.4	Conduct UAT for All Reports with Hospital Management	◇				○	◇			●
1.6.1.1	Create Role-Based User Manuals & Training Materials	○	◇				◇	◇		
1.6.1.2	Conduct Hands-On Training Sessions per Department	○	◇				○	◇		
1.6.1.3	Run System-Wide UAT & Resolve Identified Defects	◇				●	◇			
1.6.2.1	Configure Production Server Environment			○	○				○	●

1.6.2.2	Perform Go-Live Cutover & Monitor System Stability	◇	◇	○	○	○	◇	◇	●	○
1.6.2.3	Set Up Helpdesk & Provide Post-Deployment Support								○	○
1.6.3.1	Compile Program Increment Report & Lessons Learned	●	○				◇			
1.6.3.2	Obtain Formal Sign-Off from CWM Hospital Management	●	□	□	□	□	◇	□	□	●

Key

Symbol	Meaning
○	Responsible
●	Approval
◇	Support
□	Notification

IS333 Project Management Semester 1, 2026

Assignment 2 Mark Allocation Agreement

After having discussed as a group, we recommend the following mark allocation to each team member based on contribution or lack of it throughout the assignment.

Member ID	Task Contribution	Percentage of assignment 2 marks
S11230430	WBS, Introduction & Project Charter	100%
S11221203	WBS, Risk Management Plan & Change Management Plan	100%
S11234883	WBS, Scheduling & Cost Management	100%
S11230686	WBS, Communication Plan & Conclusion	100%

Signed

Member name	ID	Signature
<u>Vishant Kumar</u>	<u>S11230430</u>	<u><i>vishant</i></u>
<u>Anav Chand</u>	<u>S11221203</u>	<u><i>anav</i></u>
<u>Rohan Nandan</u>	<u>S11234883</u>	<u><i>rohan</i></u>
<u>Zhixian Chen</u>	<u>S11230686</u>	<u><i>zhixian</i></u>